

SIG731 : Task 5C

Tableau or PowerBI Dashboard

1 Introduction

Tasks 5–8 are not obligatory; you can submit them in any order (or decide not to tackle them at all). C/D/HD is merely a subjective estimate of their difficulty level. For each task that you successfully complete, you score 10 points (and for those that are not 100% correct, no points will be given).

The due date for submission is **Week 6 (Sunday)**. After this due day, the evaluation team will not provide any feedback for your work in Olympus. The submission due day is **9th February (Sunday)**. Start tackling this task as early as possible. If we find your first solution incomplete or otherwise incorrect, you will still be able to amend it based on the generous feedback we will give you (allow 3–5 working days). In case of any problems/questions, do not hesitate to attend our on-campus/online classes or use the Discussion Board on the unit site.

Submitting after the aforementioned due date might incur a late penalty. The **cut-off date is Week 7 (Sunday)**. There will be **no extensions** and no solutions will be accepted thereafter. At that time, if your submission is not 100% complete, it will be marked as FAIL, without the possibility of correcting and resubmitting. To ensure a fair environment for all, we are always very strict about deadlines.

2 Task

Your aim is to create one data visualisation **dashboard** that can be viewed by the marking tutors through a web browser: either using *Tableau Public* **or** one using *Power BI*.

Make sure that the dashboard includes your name and student ID.

Reproduce the analysis that you have performed when solving task 4P, i.e., of the `nycflights13_weather.csv.gz` (manually decompress the file first) dataset, which gives the hourly meteorological data for three airports in New York: LGA, JFK, and EWR for the whole year of 2013.

This includes:

1. Converting all columns so that they use metric (International System of Units, SI) or derived units (replace old or add new columns).

All data transformations **must** be done using *PowerBI or Tableau*! **Do not pre-process the data sets in pandas.**

2. Computing daily mean wind speeds for the LGA airport.
3. Presenting the daily mean wind speeds at LGA on a plot (a line graph, whatever is more readable).
4. Identify the 10 windiest days at LGA. Present them in a table (dates and the corresponding total wind speeds).

Make sure your results match the reference ones (from the Task 4 sheet).

5. Similarly as in Task 4P, additionally visualise the monthly mean wind speeds at the three airports (on a single plot). Remember to remove the outlier!

3 Artefacts

You should submit a single **PDF** document (you can create it in any program, including Jupyter).

At the start of the document, you need to provide: the task **title** (e.g., *Task 42: How Much I Love This Unit*), your **name**, **student number** and **email address**.

Please provide the **URLs** and *clickable* **links** to your dashboard. Make sure they can be accessed publicly, without requesting any special permissions.

Then, include 3–5 **screenshots** of the dashboard.

4 Intended Learning Outcomes

ULO	Is Related?
ULO1 (Data Processing/Wrangling)	YES
ULO2 (Data Discovery/Extraction)	YES
ULO3 (Requirement Analysis/Data Sources)	YES
ULO4 (Exploratory Data Analysis)	YES
ULO5 (Data Privacy and Ethics)	